

Stem cell-specific activation of an ancestral *myc* protooncogene with conserved basic functions in the early metazoan *Hydra*

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The *c-myc* protooncogene encodes a transcription factor (Myc) with oncogenic potential. Myc and its dimerization partner Max are bHLH-Zip DNA binding proteins controlling fundamental cellular processes. Deregulation of *c-myc* leads to tumorigenesis and is a hallmark of many human cancers. We have identified and extensively characterized ancestral forms of *myc* and *max* genes from the early diploblastic cnidarian *Hydra*, the most primitive metazoan organism employed so far for the structural, functional, and evolutionary analysis of these genes. *Hydra myc* is specifically activated in all stem cells and nematoblast nests which represent the rapidly proliferating cell types of the interstitial stem cell system and in proliferating gland cells. In terminally differentiated nerve cells, nematocytes, or epithelial cells, *myc* expression is not detectable by *in situ* hybridization. *Hydra max* exhibits a similar expression pattern in interstitial cell clusters. The ancestral *Hydra Myc* and Max proteins display the principal design of their vertebrate derivatives, with the highest degree of sequence identities confined to the bHLH-Zip domains. Furthermore, the 314-amino acid *Hydra Myc* protein contains basic forms of the essential Myc boxes I through III. A recombinant *Hydra Myc/Max* complex binds to the consensus DNA sequence CACGTG with high affinity. Hybrid proteins composed of segments from the retroviral v-Myc oncoprotein and the *Hydra Myc* protein display oncogenic potential in cell transformation assays. Our results suggest that the principal functions of the Myc master regulator arose very early in metazoan evolution, allowing their dissection in a simple model organism showing regenerative ability but no senescence.

cell proliferation | cnidaria | development | transcription factor

The *myc* oncogene was originally identified as a highly oncogenic retroviral allele (*v-myc*), derived from the cellular *c-myc* protooncogene by transduction (1). The protein product (Myc) of *c-myc* represents the central part of a transcriptional regulator network controlling the expression of up to 15% of all human genes and regulating fundamental cellular processes like growth, proliferation, differentiation, metabolism, and apoptosis (2, 3). Deregulation of *c-myc* leading to elevated levels of Myc is a frequent event in tumorigenesis, occurring in about 30% of all human cancers (4, 5). Myc is a bHLH-Zip protein containing protein dimerization domains (helix-loop-helix, leucine zipper) and a DNA contact surface (basic region) (2). Myc forms heterodimers with the bHLH-Zip protein Max, binds to specific DNA sequence elements (E-boxes), and is part of a transcription factor network including additional bHLH-Zip proteins (2). Myc-Max heterodimers are implicated in transcriptional activation of distinct target genes, but Myc has also been associated with transcriptional repression (2, 3, 6). Many of the genes activated by Myc are related to processes of cell growth and metabolism, including protein synthesis, ribosomal biogenesis, glycolysis, mitochondrial function, and cell cycle progression (3, 7, 8). Most of the genes repressed by Myc are involved in cell cycle arrest, cell adhesion, and cell-to-cell communication (3, 7, 8). Invertebrate orthologs of Myc and Max proteins have been identified in the fruit fly

Drosophila melanogaster, a triploblastic bilaterian model organism (9, 10). *Drosophila Myc* (dMyc) controls cell growth and cell size (11) and regulates cell competition in a dose-dependent manner (12). dMyc and dMax bind to a large number of genomic E-boxes to regulate the expression of many genes including key regulators of ribosome biogenesis (13, 14).

The freshwater polyp *Hydra* is a classic diploblastic model system to study pattern formation, regeneration, and stem cell dynamics in an evolutionary context (15–19), and belongs to the animal phylum cnidaria that branched off almost 600 million years ago. Its simple body plan is composed of three independent cell lineages. Two lineages of epithelial muscle cells form unicellular sheets (ectoderm, endoderm), shape the body of the polyp, and carry its morphogenetic capacities. The third lineage is a stem cell system dispersed in the interstitial spaces between epithelial muscle cells. The multipotent stem cells are located in the ectoderm of the gastric region. They give rise to ectodermal and endodermal differentiation products that are essential for the polyp's behavior (nerve cells, nematocytes, and gland cells) and sexual reproduction (gametes) (20). Ectodermal and endodermal epithelial muscle cells proliferate with a cell cycle length of about 3.5 d, coincident with the polyp growth rate (21). The interstitial stem cells proliferate much faster with a cell cycle length of about 1 d (22). However, only 60% of the daughter cells remain stem cells, whereas 40% differentiate into nerve cells, nematocytes, or gland cells (23, 24).

Here, we describe the cloning and structural and functional analysis of *myc* and *max* genes from *Hydra*, representing the oldest and most primitive metazoan organism used so far for molecular analyses of these genes and their protein products. Intriguingly, the basic biochemical and biological functions defined for vertebrate Myc and Max proteins are already laid out in their early ancestral homologs. Furthermore, the specific expression of *Hydra myc* and *max* in all rapidly proliferating cell types of the interstitial stem cell system points to an early role in fundamental cellular regulation, rendering *Hydra* an ideal model system for biochemical and genetic analyses of these important regulatory genes and their pleiotropic impact on growth and development.

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Results

Identification and Cloning of *Hydra* *myc* and *max* Genes. Starting from the *Hydra* EST (expressed sequence tags) database containing entries related to vertebrate *myc* and *max* genes, a bioinformatic search of the emerging *Hydra magnipapillata* genome database led to the identification of open reading frames encoding putative *Hydra* Myc and Max proteins. A recent search using the most current release of the *Hydra* genome database revealed the presence of additional *myc*-related sequences in the completed genome sequence. Two deduced protein sequences (termed Myc1 and Myc2) display the principal topography of Myc proteins, one (Myc1) encoded by the gene initially cloned in this study and now termed *myc1*. Comparison of the deduced protein sequences with those of the human or chicken c-Myc and Max homologs revealed overall sequence identities of 31.5% or 27.4% for the Myc1 protein, and 55.0% or 54.4% for the Max protein, respectively (Fig. 1). The highest degree of sequence identity was found in the bHLH-Zip regions of Myc1 (44.7% identity) and Max (81.0% identity), necessary for dimerization and sequence-specific DNA binding (Fig. 1). The amino-terminal *Hydra* Myc1 domain exhibits less overall sequence identity to the human homolog, although Myc boxes I, II, IIIa, and IIIb, essential for Myc-induced cell transformation and transcriptional regulation (25–28), display identities of 30.0%, 43.8%, 33.0%, and 41.7%, resp. The structures of *Hydra* Myc1 determined here and of the deduced Myc2 protein display the same principal topography and similar evolutionary relationship to the human protein (Fig. S14), but share only 23.6% overall identities among themselves. Notably, the exon-exon junctions of *Hydra* *myc1* and *myc2*, and of human c-*myc*, share very similar nucleotide sequences, and the amino acids around the junctions are either identical or

chemically conservative substitutes (Fig. S1B). The specific sequence identities among the *Hydra*, chicken, and human Myc and Max proteins are summarized in Fig. S1C. The location of *myc1* and *myc2* on the *Hydra* genome is shown in Fig. S1D. Intriguingly, *myc2* maps directly adjacent to the *CAD* gene, a bona fide Myc target in mammals (7, 8), encoding the multifunctional protein carbamoyl-phosphate synthetase/aspartate transcarbamoylase/dihydroorotase. Homology modeling of the dimerization/DNA binding domain of *Hydra* Myc/Max, using the 3D structure available for the corresponding region of the human proteins (29) as a template, revealed specific conservation of residues at the contact surfaces (Fig. S2). Molecular phylogenetic analyses revealed that the *Hydra* Myc and Max protein sequences branch off from basal positions within the trees (Fig. S3A and B). Also, the analyses suggest that the diversification of C-/B-, L-, and N-/S-Myc subfamilies occurred only within the vertebrate lineage (Fig. S3A).

The predicted coding regions of *Hydra* *myc1* (314 amino acids) and *max* (176 amino acids) (Fig. 1) were cloned by using total mRNAs isolated from whole *Hydra magnipapillata* animals as a template for cDNA synthesis followed by PCR. Sequence analysis of the full-length 942-bp *Hydra* *myc1* and the 528-bp *Hydra* *max* coding regions revealed that the *max* sequence was identical to the sequence deduced from the *Hydra* genome, whereas two nucleotide substitutions were found in the *myc1* coding sequence leading to one amino acid substitution (G188D) in comparison with the predicted *Hydra* Myc1 protein from the database. Furthermore, only a small fraction of the *Hydra* *myc1*-specific cDNAs encode the full-length 314-amino acid Myc1 protein, whereas the majority of cDNAs encode a shorter 289-amino acid version starting with the methionine at position 26 (Fig. 1).

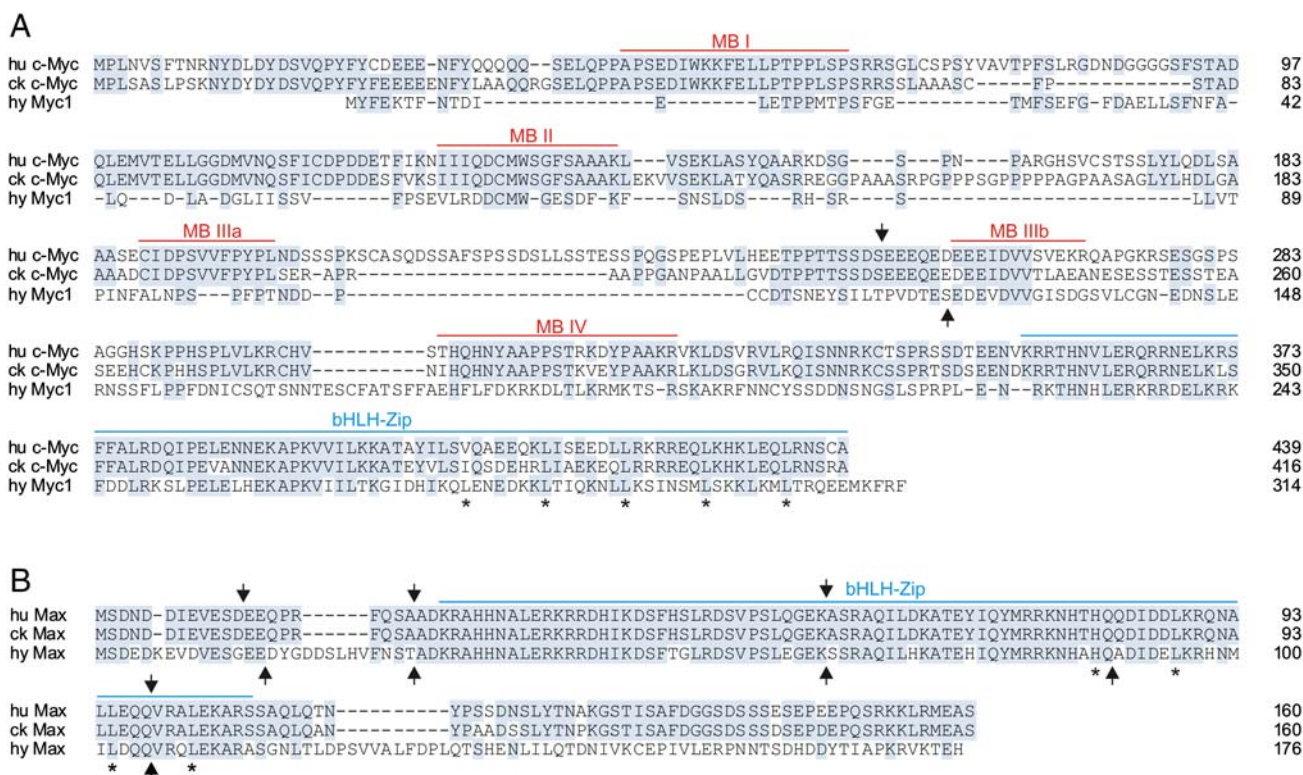


Fig. 1. Amino acid sequences of *Hydra* Myc1 and Max proteins and alignment with their vertebrate homologs. (A) Alignment of human (Hu) c-Myc, chicken (Ck) c-Myc, and *Hydra* (Hy) Myc1 sequences (GenBank accession nos.: hu c-Myc, NP_002458; ck c-Myc, NP_001026123; hy Myc1, GQ856263). (B) Alignment of human, chicken, and *Hydra* Max sequences (GenBank accession nos.: hu Max, NP_002373; ck Max, P52162; hy Max, GQ856264). Identical residues are shaded in blue, gaps are indicated by Dashes. The positions of the Myc boxes (MB) I–IV in the transactivation domains of the vertebrate proteins and of the bHLH-Zip region in the Myc or Max DNA binding domains are indicated. Conserved heptad repeat residues in the leucine zipper regions are marked by Asterisks. The positions corresponding to exon-exon junctions are indicated by Arrows above (HU, CK) and below (HY) the alignments. Alignments were generated by using the ClustalW algorithm with additional manual adjustments.

In a Northern analysis using poly(A)⁺ RNAs from whole polyps and *Hydra myc1* and *max* cDNAs as probes, single transcripts with sizes of 1.20 kb and 1.25 kb, resp., were detected, which is compatible with the sizes of the predicted open reading frames (Fig. 2A). Whereas the total expression level of *myc1* in the whole animal was rather low, *max* was abundantly expressed. Hybridization of the same filters with a *Hydra CAD* cDNA probe revealed the presence of a single 7.0-kb transcript (Fig. 2A). By using polyclonal antibodies generated against recombinant *Hydra* Myc1 and Max proteins, endogenous *Hydra* Myc1 and Max proteins were detected by immunoprecipitation or immunoblot analysis (Fig. 2B and C). For size comparison, in vitro translated full-length *Hydra* Myc1 (*M_r* 36,082) and Max (*M_r* 20,234) proteins, and the shorter Myc1 protein (*M_r* 33,174) were included in the analysis. Both in vitro translated and endogenous *Hydra* Max proteins display an apparent *M_r* of 22,000. Most of the endogenous *Hydra* Myc1 protein has an apparent *M_r* of 35,000, similar to the in vitro translated shorter 289-amino acid form (p36), whereas in vitro translated full-length *Hydra* Myc1 displays an apparent *M_r* of 39,000.

***Hydra myc1* and *max* Genes are Activated in the Interstitial Stem Cell Lineage.** In intact polyps, in situ hybridization revealed *myc1* gene expression in a large number of cells and cell nests of the interstitial stem cell lineage (Fig. 3A and Fig. S4A). Expression was restricted to the gastric region; whereas head (hypostome and tentacles) and foot (lower peduncle and basal disc) were free of *myc1* expressing cells (Fig. 3A). To analyze cell type-specific expression in more detail, we performed in situ hybridization on macerated *Hydra* single cells spread onto microscope slides, permitting unambiguous classification of cell types based on cell shape and nuclear morphology (30). We found *myc1* mRNA in all interstitial stem cells (Fig. 3C and E), in all proliferating nematoblast nests (Fig. 3D and F), and in a major fraction of gland cells (Fig. 3C and E). The differentiating and fully differentiated cell types of the nerve cell and nematocyte pathways showed no *myc1* expression. Ectodermal and endodermal epithelial cells also showed no *myc1* expression in both whole mount and maceration

preparations (Fig. 3A, C–F). Also, the *myc1* gene was not transcriptionally upregulated during morphogenetic processes such as asexual bud formation and regeneration. Taken together, *myc1* was specifically activated in all stem cells and nematoblast nests which represent the rapidly proliferating cell types of the interstitial stem cell system, and in proliferating gland cells (Fig. 3G).

In intact *Hydra*, the *max* gene exhibited an expression pattern similar to the *myc1* gene in cell clusters of the interstitial stem cell lineage (Fig. 3B and Fig. S4B). Quantitative analyses of nest sizes showed similar results for the *myc1* and *max* genes (Fig. S4C), with *myc1*- and *max*-positive nests containing 4, 8, or 16 cells. Previous analyses of genes activated in post-mitotic, differentiating nematocyte nests like *HvJNK* (31) and *Nowa* (32) had shown nest sizes between 8 and 32 cells. In addition, double in situ hybridization exhibited basically no overlap between *myc1*- and *Nowa*-expressing nests (Fig. S4D–F). Hence, *myc1* and *max* are likely coactivated in proliferating nematoblasts (Fig. 3G). At present, it is unclear to what extent *max* is upregulated in interstitial stem and gland cells. Whole mount preparations clearly showed *max* expression also in epithelial cells throughout the entire polyp (Fig. 3B and Fig. S4B). This is consistent with our results from Northern hybridization (Fig. 2A) and with the more ubiquitous activation patterns observed for *max* in various bilaterians.

Biochemical Properties of *Hydra* Myc and Max Proteins. To test if the principal biochemical functions of Myc and Max proteins have already emerged in the early metazoan *Hydra*, the coding sequences of full-length *Hydra* Max and of the carboxyl-terminal domain of *Hydra* Myc1 containing the bHLH-Zip domain (amino acid residues 201–314) (Fig. 1) were inserted into pET vectors for recombinant protein expression in *Escherichia coli*. As a control, the chicken Max p14 protein containing the bHLH-Zip region (amino acid residues 22–113) (33) was expressed analogously. The proteins *Hydra* Max p22 (*M_r* 20,234), *Hydra* Myc1 p16 (*M_r* 13,745), and chicken Max p14 (*M_r* 10,927) were efficiently expressed, and the soluble fractions were purified to homogeneity by using liquid chromatography systems (Fig. 4A). The identities of recombinant *Hydra* Myc1 p16 and Max proteins were verified by mass spectrometry (Fig. S5), fragment ion mapping (Fig. S6), and immunoblot analysis (Fig. 4A). To test if *Hydra* Myc1 p16 and *Hydra* Max p22 proteins show DNA binding activity, similar to that of chicken Max p14 (33), electrophoretic mobility shift analyses (EMSA) were performed. The analysis showed that *Hydra* Myc1 p16 binds to double-stranded DNA containing a consensus Myc binding site with high efficiency, whereas *Hydra* Max p22 displayed lower binding affinity (Fig. 4B). The apparent size of the *Hydra* Myc1 p16-DNA complex is higher than expected possibly due to binding of higher order Myc oligomers as proposed previously (29, 34). In agreement with results obtained with chicken proteins (33), the DNA affinity is enhanced when an equimolar mixture of *Hydra* Myc1 p16 and *Hydra* Max is used, allowing the formation of heterodimeric protein complexes (Fig. 4B). The presence of both proteins in these complexes was verified by coinubation with antibodies directed against *Hydra* Myc1 or Max leading to partial supershift (α -hy Max) or inhibition (α -hy Myc1) of DNA binding. To quantify the *Hydra* Myc1 or Max DNA binding affinities, increasing amounts of proteins were added to constant amounts of DNA in EMSA analyses (Fig. S7A), the ratios of bound to total DNA were determined, and the dissociation constants (*K_d*) for the protein-DNA complexes were calculated (Fig. S7B). The *K_d* values for protein-DNA complexes formed by *Hydra* Myc1 p16/*Hydra* Max p22, *Hydra* Myc1 p16, and *Hydra* Max p22 were determined to 1.7×10^{-8} M, 3.2×10^{-7} M, and 1.0×10^{-6} M, resp. Compared with the *K_d* for the avian Myc/Max heterodimer binding to the same DNA (33), the DNA affinity of *Hydra* Myc1/Max is about 100-fold lower. Specific DNA binding activity could also be demonstrated for the nearly full-length *Hydra* Myc1 p36

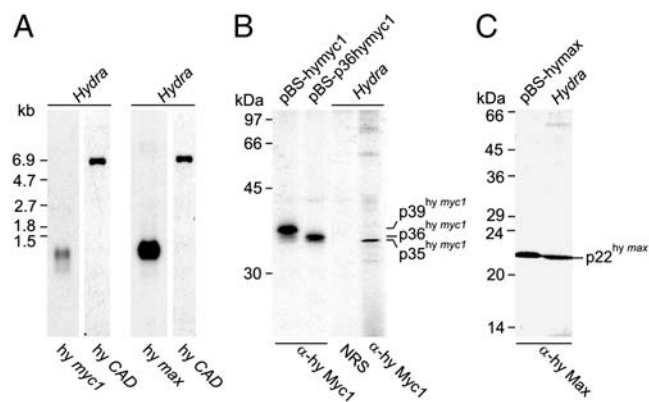


Fig. 2. Expression of *Hydra myc1* and *max* genes and their protein products. (A) Northern analyses using aliquots (2.0 μ g) of poly(A)⁺-selected RNAs from whole *Hydra* animals and *Hydra myc1* or *max* specific cDNA probes. Each filter was stripped and rehybridized with a *Hydra CAD* specific cDNA probe. (B) Immunoprecipitation analysis using aliquots (5×10^6 cpm) of boiled cell extracts from [³⁵S]methionine-labeled *Hydra* animals and a polyclonal antiserum directed against *Hydra* Myc1 recombinant protein (α -hy Myc1), or normal rabbit serum (NRS). For comparison, [³⁵S]methionine-labeled *Hydra* Myc1 p39 and p36 proteins were also produced by in vitro translation of corresponding cDNAs cloned in Bluescript (BS) vectors and immunoprecipitated. Proteins were analyzed by SDS/PAGE (10%, wt/vol). (C) Immunoblot analysis of a 30- μ g aliquot of total cell proteins from whole *Hydra* polyps and of in vitro translated *Hydra* Max using a polyclonal antiserum directed against *Hydra* Max recombinant protein (α -hy Max). Proteins were resolved by SDS/PAGE (12.5%, wt/vol).

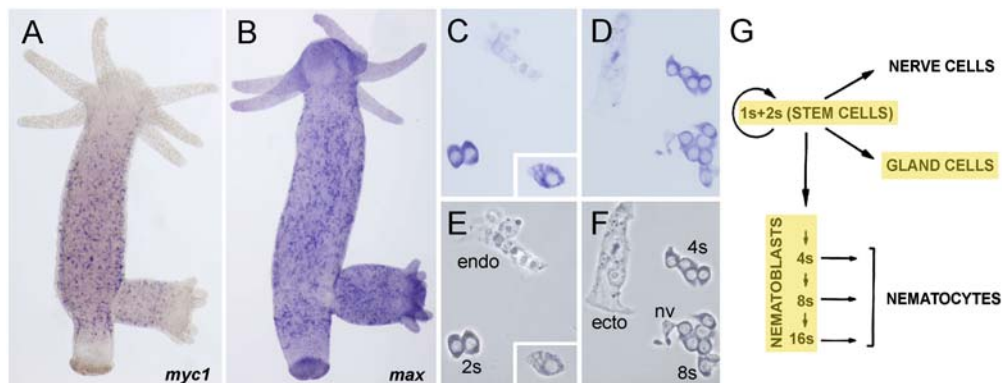


Fig. 3. Expression patterns of *Hydra myc1* and *max* visualized by in situ hybridization. (A, B) *myc1* and *max* are activated in cells belonging to the interstitial stem cell lineage in the gastric region of intact, budding polyps. In addition, *max* is expressed at a lower level in the epithelium throughout the entire body column. (C–F) *myc1* expression in interstitial stem cells (C and E), proliferating nematoblast nests (D and F), and a gland cell (Insert in C and E), visualized by in situ hybridization in macerated single cell preparations. (C and D) bright field optics; (E and F) phase contrast optics. Ecto: ectodermal epithelial cell; Endo: endodermal epithelial cell; and NV: nerve cell. (G) Scheme of the differentiation pathways in the interstitial stem cell system with *myc1* expressing subpopulations shaded in Yellow.

recombinant protein, following solubilization from inclusion bodies and renaturation (Fig. S7C).

Hydra-Viral Myc Hybrid Proteins Induce Cell Transformation. To explore if some of the principal biological functions of vertebrate Myc are already associated with their ancestral counterparts, the *Hydra myc1* and *max* coding regions, and hybrids between *Hydra* and viral *myc* were inserted into the replication-competent retroviral RCAS vector and tested for their potential to induce cell transformation in avian fibroblasts. In the hybrid constructs, the amino-terminal transcriptional regulation and carboxyl-terminal DNA binding domains had been mutually exchanged (hy/v-myc, v/hy-myc) (Fig. S5A and Fig. S8A). The empty RCAS vector and the RCAS-v-myc construct encoding the 416-amino acid viral Myc (v-Myc) protein were used as controls. The identities of the DNA inserts were verified by sequencing and by in vitro translation and immunoprecipitation of the protein products by using specific antibodies recognizing the *Hydra* Myc1 p39, *Hydra* Max p22, and avian v-Myc proteins (Fig. S8B). The in vitro translated hybrid proteins hy/vMyc (M_r 34,910) and v/hy-Myc (M_r 47,210), and the v-Myc protein (M_r 46,095) display an apparent M_r of 37,000, 53,000, or 52,000, resp. (Fig. S8B). The retroviral constructs were transfected into primary quail embryo fibroblasts (QEF), and cells were passaged several times. Expression of ectopic viral and hybrid Myc proteins and of the endogenous c-Myc protein comigrating with the v-Myc protein was monitored by immunoprecipitation analysis (Fig. S5B). The c-Myc protein is not detectable in cells that contain high levels of v-Myc or v/hy-Myc, which is due to negative transcriptional regulation of c-myc by v-Myc (35). Ectopic expression of *Hydra* Max did not cause any significant cellular alterations, and *Hydra* Myc1 induced only a marginal increase in cell proliferation. In contrast, expression of both hybrid proteins, hy/v-Myc and v/hy-Myc, led to cell transformation manifested by focus formation, enhanced proliferation, and anchorage-independent growth (Fig. S5C and D). Transformation by both hybrid proteins was less efficient than that induced by the authentic v-Myc oncoprotein. However, these results show that both the amino-terminal and the carboxyl-terminal domain of *Hydra* Myc1 are principally capable to substitute for the corresponding regions in the highly oncogenic retroviral v-Myc protein.

Discussion

Converse transcriptional regulation of growth promoting or suppressing targets is implicated in Myc-induced oncogenesis (3, 6–8). The identification of such genes relevant for normal develop-

ment, tumorigenesis, or both, is severely complicated by the large number of direct or indirect Myc targets identified by expression profiling or genomic DNA binding (3). Furthermore, the possible role of Myc as a coactivator or corepressor of other transcription factors, and the recent evidence that many biological functions of Myc are independent of Max or even E-box binding amplify the range of possible Myc targets (3, 36).

Identification and analysis of *myc* and *max* genes and proteins from genetically tractable invertebrate organisms like *Drosophila* substantially enhanced the analysis of the Myc/Max network (10, 36). Hence, characterization of these genes from yet much simpler pre-bilaterian organisms at the base of metazoan evolution may provide further insight into the basic biochemistry and biology of these important cellular regulators. Using in silico genomic analyses, the presence of *myc*-like or *myc*-related genes has been predicted for all sequenced genomes of metazoans with the exception of the nematode *Caenorhabditis elegans*, where the lack of *myc* genes is due to secondary loss (37, 38). Representing the sister group to the metazoan clade, the choanoflagellate *Monosiga brevicollis* is the most simple organism for which a protein distantly related to metazoan Myc proteins has been predicted from the genome sequence (37, 39). We have identified, cloned, and extensively characterized authentic *myc* and *max* genes from a diploblastic cnidarian organism, *Hydra magnipapillata*, that branched off in metazoan evolution almost 600 million years ago. The *Hydra* Myc1 protein shares the principal topography with vertebrate Myc proteins. A direct comparison reveals short conserved amino acid motifs in the N-terminal Myc boxes I, II, IIIa, and IIIb. Hence, this part of the protein evolved in the common cnidarian-bilaterian ancestor. The C-terminal bHLH-Zip domain represents an ancient class of eukaryotic DNA binding domains, with high sequence identity and structural homology between *Hydra* Myc1 and vertebrate Myc proteins (Fig. 1 and Fig. S2). Notably, the position of the exon-exon junction that is fixed across bilaterian *myc* genes (10) is also conserved in the pre-bilaterian *Hydra myc1* gene (Fig. S1). The *Hydra* Myc1 protein dimerizes with *Hydra* Max and binds to E-box DNA with high affinity, indicating a very early origin of these basic biochemical properties. Intriguingly, oncogenic activity of truncated mutant versions of the retroviral v-Myc protein can, at least partially, be rescued by the corresponding *Hydra* Myc1 protein segments. Hence, even basic biological functions of vertebrate Myc proteins emerged very early in their ancestral homologs.

Recent database updates reveal four *myc*-related genes in the *Hydra* genome. Two of these genes, *myc1* isolated here, and *myc2*, are clear *myc* orthologs displaying structural conservation in their

ribosome biogenesis. Under phase contrast optics (Fig. 3*E* and *F*), *myc1*-expressing interstitial stem cells, nematoblasts, and gland cells show a characteristic, dark cytoplasm based on the presence of very large numbers of ribosomes (30). Furthermore, a recent *in silico* analysis of metazoan and non-metazoan genomes has shown that Myc- and Max-specific bHLH-Zip DNA binding domains coevolved with the appearance of E-box sites in the core promoters of nearly all genes involved in ribosome biogenesis (38). The principal structural and functional similarities between *Hydra* Myc and Max proteins and their vertebrate derivatives provide the basis for using this simple organism near the base of metazoan evolution as a model system for cellular regulation by these proteins.

Materials and Methods

Animals. *Hydra vulgaris* strain Basel and *Hydra magnipapillata* wildtype strain 105 were used in this study. Mass cultures were kept at 18 °C and fed daily with freshly hatched *Artemia nauplii*. Experimental animals were collected 24 h after the last feeding (19).

DNA Cloning and Nucleic Acid Analysis. Molecular cloning, DNA sequencing, and Northern analysis have been described (6, 40). For RNA isolation, ~1,500 polyps of *Hydra magnipapillata* were lysed in 50 mL of a buffer containing 4 M guanidine thiocyanate, 25 mM sodium acetate pH 6.0, and 0.835% (vol/vol) 2-mercaptoethanol. Total RNA was isolated by CsCl density centrifugation, and mRNA was enriched by poly(A)⁺-RNA selection as described (40). Cloning of the coding regions of the predicted *Hydra myc* and *max* genes by cDNA synthesis and PCR is described in *SI Text*.

In situ Hybridization. Whole mount *in situ* hybridization with digoxigenin-labeled RNA probes was done as described (41). For *in situ* hybridization in single cell preparations, *Hydra* were macerated for 20 min (30). Cell suspensions were fixed in 4% paraformaldehyde for 30 min, spread onto microscope slides, and dried briefly. Hybridization was as described (42).

Cells and Retroviruses. Cell culture, DNA transfection, and transformation assays of quail embryo fibroblasts (QEF) were performed as described (6, 40, 43). The construction of retroviral and Bluescript vectors carrying *Hydra myc* and *max* genes or *Hydra*-viral hybrid genes is described in *SI Text*.

Protein Chemistry. *In vitro* translation, immunoprecipitation of L-[³⁵S]methionine-labeled proteins, SDS/PAGE, and immunoblotting were done as described (6, 40). Expression and purification of recombinant proteins was performed essentially as described previously (33). Construction of pET expression vectors carrying *Hydra myc* and *max* gene sequences, mass spectrometry and sequencing of recombinant proteins, generation of polyclonal antisera directed against *Hydra* Myc and Max recombinant proteins, and metabolic labeling of *Hydra* proteins are described in *SI Text*.

EMSA. EMSA analysis was performed as described (33, 35). DNA binding reactions (20 µL) were performed at 25 °C for 45 min in a buffer containing 10 mM Tris HCl pH 7.5, 0.5 mM EDTA, 65 mM KCl, 5 mM MgCl₂, 1 mM DTT, 100 µg/mL BSA, 10% (vol/vol) glycerol. For supershift analysis, polyclonal antisera (0.5 µL) were added after 15 min. Protein-DNA complexes were resolved by native 6% (wt/vol) PAGE, and radioactive signals were quantified by using a PhosphorImager (Molecular Dynamics).

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